

BEFORE

Some Pacific lands were settled many millennia before they were charted by Europeans in the late 18th century.

AUSTRALIA AND NEW ZEALAND

The **Maori**, from Polynesia, settled New Zealand by about 1400 <<216–17. But arid Australia remained uninhabited except to the **Aborigines** who landed there 40,000 years earlier.

UNCHARTED TERRITORY

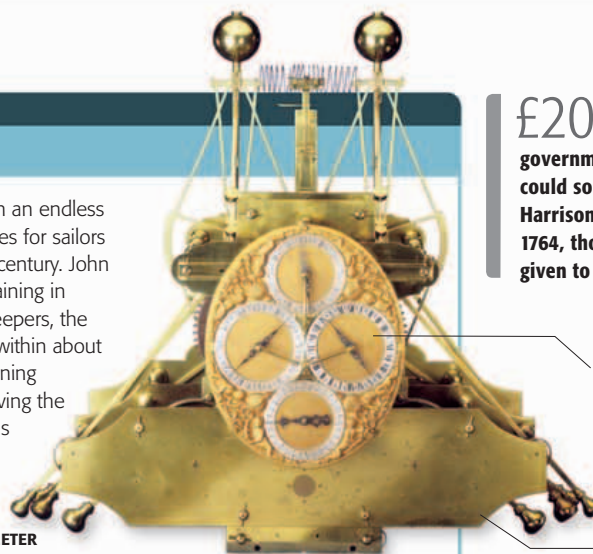
To Europeans, *Terra Australis Incognita* (Unknown Southern Continent) was a hypothetical landmass somewhere in the Pacific Ocean 320–21 >>.

INVENTION

HARRISON'S CHRONOMETER

The problem of determining longitude on an endless horizon was one of the greatest challenges for sailors navigating the Pacific Ocean in the 18th century. John Harrison (1693–1776) had no formal training in clock-making but built five marine timekeepers, the best of which could keep time at sea to within about one second per day—a staggering positioning accuracy of about 1,600 ft (500 m)—solving the navigational problem. This huge clock was his first prototype, called H1. It measures about 3 ft x 3 ft (1 m x 1 m) and weighs 72 lb (33 kg). It performed admirably at sea.

NAUTICAL CHRONOMETER



£20,000 The prize offered by the British government in 1714 to the person who could solve the longitude problem. John Harrison eventually won the prize in 1764, though the full amount was not given to him until 1773.

uses 24-hour dial; runs without pendulum or any lubrication

two 5-lb (2-kg) weights counteract the movement of a rolling ship

Exploring the Pacific

Less than 250 years ago the Pacific Ocean remained the last great unknown area on Earth, at least to European explorers and traders. European superpowers were rapidly exploring the rest of the world from the 15th century onward, yet the Pacific Ocean remained a mystery for another 400 years.

By the middle of the 18th century, Europeans had mapped the coastlines of North and South America, Asia, and Africa, and vast empires were well underway in these areas. So it may seem odd that the Europeans did not attempt to colonize the islands of the Pacific Ocean until the late 18th century.

The problem of the Pacific

Simply put, there was nothing to lure the Europeans to the Pacific—no tales of El Dorado, no indigenous empires to conquer and convert, and no lucrative trade to exploit. There were also geographical, technological, and logistical barriers: the Pacific covers an area larger than all the continents'

landmass combined, and the prevailing winds also narrowed the sea routes available. Earlier Polynesian voyagers—the Maori who had sailed to New Zealand and the discoverers of the Hawaiian Islands (see pp.216–17)—had steered their canoes by means of the sun and the stars, and patterns in the ocean's currents and waves. European sailors had effective sextants, compasses, and other navigational aids, but they still could not determine longitude. This is the measurement of the position of a point east or west of the prime meridian (where longitude is 0°). It can be calculated by comparing the time at any point to the time at the prime meridian, but to know the time at Greenwich (the British meridian) after

a voyage to the other side of the world required an extremely accurate clock. The ability to determine longitude was vital—it would allow sailors to chart their precise location and that of the islands they visited. The Solomon Islands, for example, were sighted and reported in 1568 but were not

Maori war canoe

The Maoris had giant war canoes (*waka taua*), which carried 70–140 armed warriors. Led by a Maori chief (*rangatira*), a war canoe had a hull carved out of a single tree trunk and an intricately carved prow.

